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Resonant tunneling diodes and manufacturing methods thereof

Abstract

The present disclosure provides a resonant tunneling diode including: a first barrier layer; a second barrier layer; a potential well layer between the first barrier layer and the second barrier layer, materials of the first barrier layer, the second barrier layer, and the potential well layer including a group III nitride, a material of the potential well layer including a gallium element; a first barrier layer between the first barrier layer and the potential well layer; and/or a second barrier layer between the second barrier layer and the potential well layer.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application is a US National Phase of a PCT Application No. PCT/CN2021/079266, filed

on Mar. 5, 2021, the contents of which are incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to the field of semiconductor technology, and more particularly to resonant tunneling diodes and manufacturing methods thereof.

BACKGROUND

(3) Terahertz technology, as a new science and technology, has important application prospects in security detection, material identification, secure communication, space exploration, high-precision radar, tissue biopsy, transient spectrum research and so on. A resonant tunneling diode, as a dual-terminal device, can produce negative differential resistance by using resonant tunneling phenomenon, which is configured to prepare terahertz radiation source, resulting in a wide range of concerns.

(4) At present, a mature resonant tunneling diode (RTD) is mainly a GaAs-based material, due to inherent performance limitations of GaAs-based materials, a power of RTD merely reaches an order of micro watts. GaN-based resonant tunneling diodes have advantages of group III nitride semiconductor materials, such as, high electron rate, high breakdown field strength, large band gap adjustable range and radiation resistance. With this configuration, it is expected to realize terahertz high-power emission at room temperature.

(5) However, at present, nitride-based resonant tunneling diodes face problems of poor device stability and low peak-to-valley current ratio, which greatly limits its practical application.

SUMMARY

(6) An object of the present disclosure is to provide a resonant tunneling diode and a manufacturing method thereof, so as to improve a device stability and a peak-to-valley current ratio.

(7) To achieve the above object, a first aspect of the present disclosure provides a resonant tunneling diode, including:

(8) a first barrier layer; a second barrier layer; a potential well layer between the first barrier layer and the second barrier layer, where a material of the first barrier layer, a material of the second barrier layer, and a material of the potential well layer all include a group III nitride, the material of the potential well layer includes a gallium element; a first isolation layer between the first barrier layer and the potential well layer; and/or a second isolation layer between the second barrier layer and the potential well layer.

(9) In some embodiments, the material of the first barrier layer and the material of the second barrier layer include at least one of AlGaN, AlInGaN, or InAlN.

(10) In some embodiments, a material of the first isolation layer and/or a material of the second isolation layer includes AlN.

(11) In some embodiments, a thickness of the first isolation layer ranges from 0.1 nm to 2 nm; and/or a thickness of the second isolation layer ranges from 0.1 nm to 2 nm.

(12) In some embodiments, the resonant tunneling diode further includes a collector electrode and an emitter electrode, the collector electrode being close to the first barrier layer, the emitter electrode being close to the second barrier layer; a third isolation layer between the collector electrode and the first barrier layer; and/or a fourth isolation layer between the emitter electrode and the second barrier layer.

(13) In some embodiments, a material of the collector electrode and a material of the emitter electrode include a GaN-based material.

(14) In some embodiments, a material of the third isolation layer and/or a material of the fourth isolation layer includes AlN.

(15) A second aspect of the present disclosure provides a method of manufacturing a resonant tunneling diode, including: epitaxially growing a first barrier layer, a potential well layer, and a second barrier layer in sequence on a substrate, where a material of the first barrier layer, a material of the second barrier layer, and a material of the potential well layer include a group III nitride, the material of the potential well layer includes a gallium element; removing the substrate; before

epitaxially growing the potential well layer, epitaxially growing a first isolation layer on the first barrier layer; and/or before epitaxially growing the second barrier layer, epitaxially growing a second isolation layer on the potential well layer.
(16) In some embodiments, the material of the first barrier layer and the material of the second barrier layer include at least one of AlGaN, AlInGaN, or InAlN.
(17) In some embodiments, a material of the first isolation layer and/or a material of the second isolation layer includes AlN.
(18) In some embodiments, a thickness of the first isolation layer ranges from 0.1 nm to 2 nm; and/or a thickness of the second isolation layer ranges from 0.1 nm to 2 nm.
(19) In some embodiments, the method of manufacturing the resonant tunneling diode further includes: epitaxially growing a collector electrode on a side of the first barrier layer away from the potential well layer; epitaxially growing an emitter electrode on a side of the second barrier layer away from the potential well layer, where a material of the collector electrode and a material of the emitter electrode both include a group III nitride;
(20) or before epitaxially growing the first barrier layer, epitaxially growing the collector electrode on the substrate, epitaxially growing the emitter electrode on the second barrier layer, where the material of the collector electrode and the material of the emitter electrode both include a group III nitride.
(21) In some embodiments, the method of manufacturing the resonant tunneling diode further includes: before epitaxially growing the collector electrode on the side of the first barrier layer away from the potential well layer, epitaxially growing a third isolation layer on the side of the first barrier layer away from the potential well layer; or after epitaxially growing the collector electrode on the substrate, and before epitaxially growing the first barrier layer, epitaxially growing the third isolation layer;
(22) and/or, before epitaxially growing the emitter electrode on the side of the second barrier layer away from the potential well layer, epitaxially growing a fourth isolation layer on the side of the second barrier layer away from the potential well layer.
(23) In some embodiments, the material of the collector electrode and the material of the emitter electrode include a GaN-based material.
(24) In some embodiments, a material of the third isolation layer and/or a material of the fourth isolation layer includes AlN.
(25) Compared with the related art, the present disclosure has the following beneficial effects: using the first isolation layer and the second isolation layer can prevent gallium atoms in the potential well layer from diffusing to the first barrier layer and the second barrier layer, ensuring that compositions of the first barrier layer and the second barrier layer are uniform, preventing an effective thickness from thinning, thereby improving a device stability and a peak-to-valley current ratio.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a cross-sectional schematic view of a resonant tunneling diode according to a first embodiment of the present disclosure;
(2) FIG. **2** is a flowchart of a method of manufacturing a resonant tunneling diode in FIG. **1**;
(3) FIG. **3** is a schematic view illustrating an intermediate structure corresponding to processes in FIG. **2**;
(4) FIG. **4** is a cross-sectional schematic view of a resonant tunneling diode according to a second embodiment of the present disclosure;
(5) FIG. **5** is a cross-sectional schematic view of a resonant tunneling diode according to a third

embodiment of the present disclosure;
(6) FIG. **6** is a cross-sectional schematic view of a resonant tunneling diode according to a fourth embodiment of the present disclosure; and
(7) FIG. **7** is a cross-sectional schematic view of a resonant tunneling diode according to a fifth embodiment of the present disclosure.
(8) To facilitate understanding of the present disclosure, all reference numerals appearing in the present disclosure are listed below:
(9) TABLE-US-00001 resonant tunneling diodes 1, 2, 3, 4, 5 substrate 10; first barrier layer 11 second barrier layer 12 potential well layer 13 first isolation layer 14 second isolation layer 15 third isolation layer 16 fourth isolation layer 17 collector electrode 20 emitter electrode 30
DETAILED DESCRIPTION OF THE EMBODIMENTS
(10) To make the above-mentioned objects, features and advantages of the present disclosure more obvious and understandable, embodiments of the present disclosure will be described in detail below with reference to the accompanying drawings.
(11) FIG. **1** is a cross-sectional schematic view of a resonant tunneling diode according to a first embodiment of the present disclosure.
(12) Referring to FIG. **1**, the resonant tunneling diode **1** includes:
(13) a first barrier layer **11**; a second barrier layer **12**; a potential well layer **13** between the first barrier layer **11** and the second barrier layer **12**, where a material of the first barrier layer **11**, a material of the second barrier layer **12**, and a material of the potential well layer **13** all include a group III nitride, and the material of the potential well layer **13** includes a gallium (Ga) element; a first isolation layer **14** between the first barrier layer **11** and the potential well layer **13**; and a second isolation layer **15** between the second barrier layer **12** and the potential well layer **13**.
(14) The first barrier layer **11**, the potential well layer **13**, and the second barrier layer **12** form a dual barrier quantum well structure. A band gap width of the potential well layer **13** is less than a band gap width of the first barrier layer **11** and a band gap width of the second barrier layer **12**. The material of the potential well layer **13** includes GaN. The material of the first barrier layer **11** and the material of the second barrier layer **12** include at least one of AlGaN, AlInGaN, or InAlN. In some embodiments, the first barrier layer **11** and/or the second barrier layer **12** are/is a single layer structure, a material of the single layer structure includes a mixture of one or more of AlGaN, AlInGaN, or InAlN. In other embodiments, the first barrier layer **11** and/or the second barrier layer **12** are/is a stacked structure, the stacked structure includes two or three layers. When the stacked structure includes two layers, the stacked structure can include, from bottom to top, an AlGaN layer and an AlInGaN layer, an AlInGaN layer and an AlGaN layer, an InAlN layer and an AlGaN layer, an InAlN layer and an AlInGaN layer, an AlGaN layer and an InAlN layer, or an AlInGaN layer and an InAlN layer. When the stacked structure includes three layers, respective layers in the stacked structure can be any one of an AlGaN layer, an AlInGaN layer, or an InAlN layer, materials of adjacent two layers in the stacked structure are different.
(15) A GaN material is lattice-matched to an AlGaN material, an AlInGaN material, or an InAlN material. However, a band gap width of the AlGaN material varies as an aluminum composition varies; if gallium atoms in the GaN material diffuse into the AlGaN material, which can cause a resonant tunneling effect of the resonant tunneling diode **1** to be unstable, thereby resulting in a low peak-to-valley current ratio of the resonant tunneling diode **1**. For the AlInGaN material and the InAlN material, in addition to band gap widths vary as aluminum composition varies, due to an indium atom having a larger atomic radius than other atoms and indium atomic components being different, activities vary greatly; and if gallium atoms in the GaN material diffuse into the AlInGaN material and the InAlN material, which can cause the resonant tunneling effect of the resonant tunneling diode **1** to be unstable.
(16) In this embodiment, it is noted that a chemical element represents a certain material, but molar ratios of respective chemical elements in a material are not limited. For example, a GaN material

includes a gallium element and a nitrogen element, but a molar ratio of the gallium element to the nitrogen element is not limited; an AlGaN material includes aluminum, gallium and nitrogen elements, but respective molar ratios of the three elements are not limited.
(17) The arrangements of the first isolation layer **14** and the second isolation layer **15** can prevent gallium atoms in the potential well layer **13** from diffusing to the first barrier layer **11** and the second barrier layer **12**, ensuring that compositions of the first barrier layer **11** and the second barrier layer **12** are uniform, preventing an effective thickness from thinning, thereby improving a device stability and a peak-to-valley current ratio of the resonant tunneling diode **1**. A material of the first isolation layer **14** and a material of the second isolation layer **15** can include AlN.
(18) A thickness of the first isolation layer **14** can range from 0.1 nm to 2 nm. A thickness of the second isolation layer **15** can range from 0.1 nm to 2 nm.
(19) The present embodiments further provide a method of manufacturing the resonant tunneling diode of FIG. **1**. FIG. **2** is a flowchart of a manufacturing method; FIG. **3** is a schematic view illustrating an intermediate structure corresponding to processes in FIG. **2**.
(20) First, referring to step S**1** in FIG. **2** and FIG. **3**, a first barrier layer **11**, a first isolation layer **14**, and a potential well layer **13** are epitaxially grown in sequence on a substrate **10**, a material of the first barrier layer **11** and a material of the potential well layer **13** both include a group III nitride, and the material of the potential well layer **13** include a gallium element.
(21) A material of the substrate **10** can include at least one of sapphire, silicon carbide, or silicon; or at least one of sapphire, silicon carbide, or silicon, and the group III nitride material; which is not limited in this embodiment.
(22) The material of the first barrier layer **11** can include at least one of AlGaN, AlInGaN, or InAlN. In some embodiments, the first barrier layer **11** is a single layer structure, a material of the single layer structure includes a mixture of one or more of AlGaN, AlInGaN, or InAlN. In other embodiments, the first barrier layer **11** is a stacked structure, the stacked structure can include two or three layers. When the stacked structure includes two layers, the stacked structure can include, from bottom to top, an AlGaN layer and an AlInGaN layer, an AlInGaN layer and an AlGaN layer, an InAlN layer and an AlGaN layer, an InAlN layer and an AlInGaN layer, an AlGaN layer and an InAlN layer, or an AlInGaN layer and an InAlN layer. When the stacked structure includes three layers, respective layers in the stacked structure can be any one of an AlGaN layer, an AlInGaN layer, or an InAlN layer, materials of adjacent two layers in the stacked structure are different.
(23) A band gap width of the potential well layer **13** is less than a band gap width of the first barrier layer **11**. The material of the potential well layer **13** includes GaN.
(24) The arrangement of the first isolation layer **14** can prevent gallium atoms in the potential well layer **13** from diffusing to the first barrier layer **11**, ensuring that compositions of the first barrier layer **11** are uniform, preventing an effective thickness from thinning, thereby improving a device stability and a peak-to-valley current ratio of the resonant tunneling diode **1**. A material of the first isolation layer **14** can include AlN.
(25) A thickness of the first isolation layer **14** can range from 0.1 nm to 2 nm.
(26) The epitaxial growth processes of the first barrier layer **11**, the first isolation layer **14** and the potential well layer **13** can include: atomic layer deposition (ALD), chemical vapor deposition (CVD), molecular beam epitaxy (MBE), plasma enhanced chemical vapor deposition (PECVD), low pressure chemical vapor deposition (LPCVD), or metal-organic chemical vapor deposition (MOCVD), or a combination thereof.
(27) Next, referring to step S**2** in FIG. **2** and FIG. **3**, a second isolation layer **15** and a second barrier layer **12** are epitaxially grown in sequence on the potential well layer **13**, a material of the second barrier layer **12** includes a group III nitride.
(28) A band gap width of the second barrier layer **12** is greater than a band gap width of the potential well layer **13**.
(29) The material of the second barrier layer **12** can include at least one of AlGaN, AlInGaN, or

InAlN. In some embodiments, the second barrier layer **12** is a single layer structure, a material of the single layer structure includes a mixture of one or more of AlGaN, AlInGaN, or InAlN. In other embodiments, the second barrier layer **12** is a stacked structure, the stacked structure can include two or three layers. When the stacked structure includes two layers, the stacked structure can include, from bottom to top, an AlGaN layer and an AlInGaN layer, an AlInGaN layer and an AlGaN layer, an InAlN layer and an AlGaN layer, an InAlN layer and an AlInGaN layer, an AlGaN layer and an InAlN layer, or an AlInGaN layer and an InAlN layer. When the stacked structure includes three layers, respective layers in the stacked structure can be any one of an AlGaN layer, an AlInGaN layer, or an InAlN layer, materials of adjacent two layers in the stacked structure are different.
(30) The arrangement of the second isolation layer **15** can prevent gallium atoms in the potential well layer **13** from diffusing to the second barrier layer **12**, ensuring that compositions of the second barrier layer **12** are uniform, preventing an effective thickness from thinning, thereby improving a device stability and a peak-to-valley current ratio of the resonant tunneling diode **1**. A material of the second isolation layer **15** can include AlN.
(31) A thickness of the second isolation layer **15** can range from 0.1 nm to 2 nm.
(32) Epitaxial growth processes of the second isolation layer **15** and the second barrier layer **12** can refer to the epitaxial growth processes of the first barrier layer **11**, the first isolation layer **14**, and the potential well layer **13**.
(33) Thereafter, referring to step S**3** in FIG. **2**, FIG. **3** and FIG. **1**, the substrate **10** is removed.
(34) The substrate **10** can be removed by laser stripping or chemical etching.
(35) In some embodiments, the substrate **10** can be not removed. In other words, the substrate **10** remains in the resonant tunneling diode **1**.
(36) FIG. **4** is a cross-sectional schematic view of a resonant tunneling diode according to a second embodiment of the present disclosure.
(37) Referring to FIG. **4**, a resonant tunneling diode **2** and a manufacturing method thereof in the second embodiment are substantially the same as the resonant tunneling diode **1** and the manufacturing method thereof in the first embodiment, except that the second isolation layer **15** and a process of manufacturing the second isolation layer **15** are omitted.
(38) FIG. **5** is a cross-sectional schematic view of a resonant tunneling diode according to a third embodiment of the present disclosure.
(39) Referring to FIG. **5**, a resonant tunneling diode **3** and a manufacturing method thereof in the third embodiment are substantially the same as the resonant tunneling diode **1** and the manufacturing method thereof in the first embodiment, except that the first isolation layer **14** and a process of manufacturing the first isolation layer **14** are omitted.
(40) FIG. **6** is a cross-sectional schematic view of a resonant tunneling diode according to a fourth embodiment of the present disclosure.
(41) Referring to FIG. **6**, a resonant tunneling diode **4** in the fourth embodiment is substantially the same as the resonant tunneling diode **1** in the first embodiment, the resonant tunneling diode **2** in the second embodiment, the resonant tunneling diode **3** in the third embodiment, except that the resonant tunneling diode **4** further includes a collector electrode **20** and an emitter electrode **30**, the collector electrode **20** is close to the first barrier layer **11**, and the emitter electrode **30** is close to the second barrier layer **12**.
(42) A material of the collector electrode **20** and a material of the emitter electrode **30** can include a group III nitride, for example, a GaN-based material, and more particularly, GaN.
(43) Correspondingly, a manufacturing method in the present embodiment is substantially the same as the manufacturing methods in previous embodiments, except that step S**3** is followed by step S**4**: the collector electrode **20** is epitaxially grown on a side of the first barrier layer **11** away from the potential well layer **13**, the emitter electrode **30** is epitaxially grown on a side of the second barrier layer **12** away from the potential well layer **13**, and the material of the collector electrode **20** and

the material of the emitter electrode **30** include a group III nitride.
(44) In other embodiments, a step of epitaxially growing the emitter electrode **30** on the side of the second barrier layer **12** away from the potential well layer **13** can be performed between step S**2** and step S**3**.
(45) A difference between the manufacturing method in the present embodiment and the manufacturing methods in the previous embodiments is that: at step S**1**: before the first barrier layer **11** is epitaxially grown, the collector electrode **20** is epitaxially grown on the substrate **10**, the material of the collector electrode **20** includes a group III nitride; after the step S**3** or between the step S**2** and the step S**3**, epitaxially growing the emitter electrode **30** on the second barrier layer **12** is performed, where the material of the emitter electrode **30** includes a group III nitride.
(46) FIG. **7** is a cross-sectional schematic view of a resonant tunneling diode according to a fifth embodiment of the present disclosure.
(47) Referring to FIG. **7**, a resonant tunneling diode **5** in the fifth embodiment is substantially the same as the resonant tunneling diode **4** in the fourth embodiment, except that a third isolation layer **16** is disposed between the collector electrode **20** and the first barrier layer **11**, and a fourth isolation layer **17** is disposed between the emitter electrode **30** and the second barrier layer **12**.
(48) The third isolation layer **16** can prevent gallium atoms in the collector electrode **20** from diffusing to the first barrier layer **11**. The fourth isolation layer **17** can prevent gallium atoms in the emitter electrode **30** from diffusing to the second barrier layer **12**.
(49) A material of the third isolation layer **16** and a material of the fourth isolation layer **17** can include AlN.
(50) A thickness of the third isolation layer **16** can range from 0.1 nm to 2 nm. A thickness of the fourth isolation layer **17** can range from 0.1 nm to 2 nm.
(51) In some embodiments, any one of the third isolation layer **16** and the fourth isolation layer **17** can be disposed.
(52) Correspondingly, the manufacturing method in the present embodiment is substantially the same as the manufacturing methods in the previous embodiments, except that at step S**4**: before the collector electrode **20** is epitaxially grown on the side of the first barrier layer **11** away from the potential well layer **13**, the third isolation layer **16** is epitaxially grown on the side of the first barrier layer **11** away from the potential well layer **13**; or at step S**1**: after the collector electrode **20** is epitaxially grown on the substrate **10** and before the first barrier layer **11** is epitaxially grown, the third isolation layer **16** is epitaxially grown;
(53) and/or, at step S**4**: before the emitter electrode **30** is epitaxially grown on the side of the second barrier layer **12** away from the potential well layer **13**, a fourth isolation layer **17** is epitaxially grown on the side of the second barrier layer **12** away from the potential well layer **13**.
(54) Epitaxial growth processes of the third isolation layer **16** and the fourth isolation layer **17** can refer to the epitaxial growth processes of the first isolation layer **14** and the second isolation layer **15**.
(55) Although the present disclosure discloses the above contents, the present disclosure is not limited thereto. One of ordinary skill in the art can make various variants and modifications to the present disclosure without departing from the spirit and scope of the present disclosure. Therefore, the protection scope of the present disclosure should be set forth by the appended claims.

Claims

1. A resonant tunneling diode, comprising: a first barrier layer; a second barrier layer; a potential well layer between the first barrier layer and the second barrier layer; a collector electrode; an emitter electrode, the collector electrode close to the first barrier layer, the emitter electrode close to the second barrier layer; a third isolation layer entirely between the collector electrode and the first barrier layer; wherein a material of the first barrier layer, a material of the second barrier layer,

and a material of the potential well layer all comprise a group III nitride, a material of the third isolation layer comprises AlN, and the material of the potential well layer comprises a gallium element; and a first isolation layer between the first barrier layer and the potential well layer.
2. The resonant tunneling diode of claim 1, wherein the material of the first barrier layer and the material of the second barrier layer comprise at least one of AlGaN, AlInGaN, or InAlN, and a structure of one or more of the first barrier layer or the second barrier layer is a stacked structure with two layers.
3. The resonant tunneling diode of claim 1, wherein a material of the first isolation layer comprises AlN.
4. The resonant tunneling diode of claim 1, wherein a thickness of the first isolation layer ranges from 0.1 nm to 2 nm.
5. The resonant tunneling diode of claim 1, further comprising: a second isolation layer between the second barrier layer and the potential well layer; wherein a material of the second isolation layer comprises AlN, and a thickness of the second isolation layer ranges from 0.1 nm to 2 nm.
6. The resonant tunneling diode of claim 1, wherein a material of the collector electrode and a material of the emitter electrode comprise a GaN-based material.
7. The resonant tunneling diode of claim 1, further comprising: a fourth isolation layer between the emitter electrode and the second barrier layer, a material of the fourth isolation layer comprising AlN.
8. The resonant tunneling diode of claim 2, wherein the stacked structure with the two layers comprises one of: an AlGaN layer and an AlInGaN layer, an AlInGaN layer and an AlGaN layer, an InAlN layer and an AlInGaN layer, or an AlInGaN layer and an InAlN layer that are arranged from bottom to top.
9. The resonant tunneling diode of claim 2, wherein the structure of the one or more of the first barrier layer or the second barrier layer is the stacked structure with three layers, where materials of two layers that are adjacent to each other in the stacked structure are different.
10. A resonant tunneling diode, comprising: a first barrier layer; a second barrier layer; a potential well layer between the first barrier layer and the second barrier layer; a collector electrode; an emitter electrode, the collector electrode close to the first barrier layer, the emitter electrode close to the second barrier layer; a fourth isolation layer entirely between the emitter electrode and the second barrier layer; wherein a material of the first barrier layer, a material of the second barrier layer, and a material of the potential well layer all comprise a group III nitride, a material of the fourth isolation layer comprises AlN, and the material of the potential well layer comprises a gallium element; and a first isolation layer between the first barrier layer and the potential well layer.